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<tr><td rowspan="2">etana Biotech

PT ETANA BIOTECHNOLOGIES INDONESIA</td><td>**STANDARD OPERATING PROCEDURE**</td><td>Page : 1 of 8</td></tr>
<tr><td>TITLE :

PERAWATAN TERHADAP MESIN PARTICLE COUNTER

MAINTENANCE OF PARTICLE COUNTER MACHINE</td><td>SOP No. : SOP/EBI/EN-063

Revision : 00

Effective Date :

Reviewed Date :</td></tr>
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PREPARED BY :	**REVIEWED BY :**	**APPROVED BY :**	**APPROVED BY :**
WENDI RUKMANSYAH (ENGINEERING SPV)	**PURNO BUDI KISWANTO (ENGINEERING MANAGER)**	**VINO SOADUON (PROD MANAGER)**	**SHANDY (QA MANAGER)**

1. TUJUAN

PURPOSE

Memberikan petunjuk yang jelas mengenai cara perawatan terhadap Mesin *Particle Counter* pada area Produksi.

Provide clear instructions on how to maintenance for Particle Counter Machine in the production area.

2. RUANG LINGKUP

SCOPE

SOP ini digunakan oleh departement Engineering sebagai pedoman dalam melakukan perawatan Mesin *Particle Counter* pada area produksi.

This SOP is used by the Engineering department as a guideline for carrying out Particle Counter Machine maintenance in the production area.

3. TANGGUNG JAWAB

RESPONSIBILITY

3.1. Teknisi Engineering *Preventive Maintenance* bertanggung jawab :

Engineering Technician Preventive Maintenance are responsible :

3.1.1. Melakukan tugas perawatan dengan baik, sesuai dengan SOP ini.

Performing maintenance duties properly, in accordance with this SOP.

3.1.2. Melakukan perawatan sesuai jadwal yang telah ditetapkan.

Perform maintenance according to a predefined schedule.

3.1.3. Melaporkan setiap terjadi kerusakan dan perbaikan yang diperlukan kepada Supervisor Engineering.

Report any damage and repairs that are needed to the Engineering Supervisor.

3.2. Supervisor Engineering bertanggung jawab :
Engineering Supervisor are responsible :

3.2.1. Membuat jadwal *maintenance* secara berkala.
Make schedule of maintenance periodically.

3.2.2. Memastikan pelaksanaan *maintenance* sesuai dengan jadwal.
Ensure implementation of maintenance is according to the schedule.

3.2.3. Memastikan semua alat ukur yang terlibat dalam proses perawatan sesuai dengan spesifikasi yang telah ditentukan (dikalibrasi).
Ensure all measuring instruments involved in the maintenance process are in accordance with the specifications specified (calibrated).

3.2.4. Mengadakan *review* bulanan terhadap hasil kerja Teknisi *Preventive Maintenance*.
Perform a monthly review of the workings of the Preventive Maintenance Technicians.

3.2.5. Memberikan pelatihan kepada departemen lain yang terkait.
Provide training to others department which related.

3.3. Manajer Engineering bertanggung jawab :
Engineering Manager is responsible :

3.3.1. Menentukan *spare parts* pengganti apabila *spare parts* yang sesuai sudah tidak dijual lagi.
Determine replace parts if suitable spare part are no longer sold.

3.3.2. Memutuskan dan mengkoordinir tindakan yang diperlukan, apabila perlu dilakukannya penggantian, modifikasi peralatan, perubahan sistem dan menyiapkan prosedur perubahan.
Resolve and coordinate necessary actions, if necessary replacement, equipment modification, system changes and prepared change control.

3.4. Operator Mesin Produksi bertanggung Jawab untuk :
The Production Machine Operator is responsible for :

3.4.1. Melaporkan jika di areanya ditemukan kelainan-kelainan/kerusakan-kerusakan yang biasanya menghambat dalam pengoperasian mesin tersebut.
Report if the area is detected by defects / abnormalities that normally impede the operation of the machine.

3.4.2. Melakukan *daily maintenance* seperti : mengencangkan, klem-klem, *ferulle*, baut-baut yang kendor, membersihkan *body* mesin / *Main Frame* setiap saat sehingga tidak berkarat dan permukaan mesin tetap baik, membersihkan tiap-tiap mesin yang menjadi tanggung jawabnya, terutama pada bagian dalam dari kerak-kerak yang menempel.

Perform daily maintenance such as: tighten ferulle clamps loose bolts, clean the body of the machine / Main Frame at all times so that it does not rust and the surface of the machine remains good, cleaned every machine that is the responsibility, especially on the inside of the crust that is attached.

3.4.3. Memastikan kondisi alat-alat listrik dan elektronik yang merupakan bagian dari mesin yang ditempatkan pada daerah rawan air, agar selalu terhindar dari percikan air.
Ensure the condition of electrical and electronic equipment that is part of the machine placed in water-prone areas, in order to always avoid splashing water.

3.4.4. Melakukan pembuangan air kondensat pada *Air Service Regulator* pada mesin mesin yang dilengkapi dengan sistem pneumatik.
Conduct condensate water discharge at Air Service Regulator on machine equipped with pneumatic system.

3.5. Staf Kalibrasi dari Department Pemastian Mutu bertanggung Jawab melakukan kalibrasi terhadap komponen komponen alat ukur yang digunakan, misalnya *Thermometer, Pressure Gauge,Thermo Controller* dll, dan melakukan kualifikasi secara rutin.
The Calibration Staff of the Quality Assurance Department is responsible for calibrating the components of the measuring instruments used, such as Thermometer, Pressure Gauge, Thermo Controller etc., and conducting regular qualifications..

4. DEFINISI
DEFINITION

4.1. Mesin *Particle counter* bekerja dengan mendeteksi partikel-partikel yang melewati sensor dan menghitungnya berdasarkan ukuran partikel tersebut. Ini membantu dalam memastikan bahwa lingkungan atau produk tertentu memenuhi standar kebersihan yang ditetapkan.
The particle counter machine works by detecting particles that pass through the sensor and counting them based on the particle size. This helps in ensuring that a particular environment or product meets the set hygiene standards.

4.2. *Autonomous maintenance* adalah perawatan mandiri mesin yang dilakukan oleh operator mesin. Bila selama ini operator hanya dilatih untuk mengoperasikan mesin, maka sudah saatnya untuk dilatih lebih lanjut.
Autonomous maintenance is an independent engine maintenance carried out by the machine operator. If all this time the operators are only trained to operate the engine, then it's time to be further trained.

5. PETUNJUK UMUM

GENERAL INSTRUCTION

5.1. Sebelum melaksanakan *preventive maintenance*, staf engineering harus berkordinasi dengan operator terkait area tersebut untuk mempersiapkan areanya.

Before implementing preventive maintenance, engineering staff must coordinate with the operator about the area to prepare the area.

5.2. Pada saat pelaksanaan perawatan operator terkait harus mendampingi Staf *Engineering* untuk menghindari kesalahan pengoperasian saat mesin dijalankan atau operator bisa menjalani *autonomous maintenance* pada mesin tersebut.

When carrying out maintenance, the related operator must accompany Engineering to avoid operating errors when the machine is running or the operator can undergo autonomous maintenance on the machine.

5.3. Keselamatan kerja harus sesalu diperhatikan pada saat *preventive maintenance*, gunakan APD (Alat pelindung diri) yang sesuai peruntukannya pada kondisi saat *maintenance.*

Work safety must be paid attention to at the time of preventive maintenance, use PPE (Personal protective equipment) which according to allotment in condition when maintenance.

5.4. Untuk penggantian komponen kritis atau *spare part* yang berdampak langsung dengan produk apabila diganti dengan beda merek dan tipe yang berbeda tetapi satu jenis maka dibuatkan pengendalian perubahan *(change control).*

For the replacement of critical components or spare parts that have a direct impact on the product when replaced with different brands and different types but one type, make change control.

6. ALAT DAN BAHAN

TOOLS AND MATERIALS

6.1**.** **ALAT**

TOOLS

1 set tools box, pada saat melakukan *preventive maintenance*.

1 set of toolbox, when performing Preventive Maintenance.

6.2. **BAHAN**

MATERIAL

6.2.1. Ethanol 20%

Ethanol 20%

6.2.2. Kain Wipes

Wipes Fabric

7. PROSEDUR

PROCEDURE

Perawatan 3 bulanan (L2) dan 1 Tahunan (L4) pada Mesin *Particle Counter*

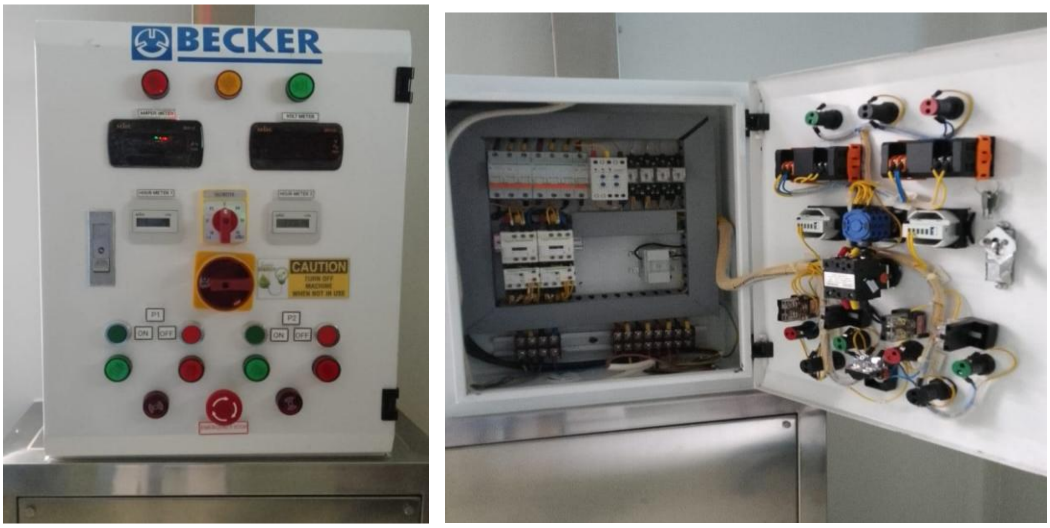
Maintenance 3 month (L2) and 1 Year (L4) of Mesin *Particle Counter Machine.*

7.1. Perawatan 3 bulanan (L2).

Maintenance 3 month (L2).

7.1.1. Perawatan pada bagian Electrical.

Maintenance of Electrical elements.



Gambar 1. Bagian panel listrik mesin particle counter untuk mesin filling bosch.

Figure 1. Particle counter machine electrical panel for bosch filling machine.

7.1.1.1. Pemeriksaan pada koneksi power listrik 3phasa dan koneksi – koneksi lainnya.

Check electrical power connections 3 Phase and other connections.

7.1.1.2. Periksa tombol - tombol pengoperasian pada panel.

Check the operating buttons on the panel.

7.1.1.3. Bersihkan panel dari debu menggunakan kuas.

Clean the panel from dust using a brush.

7.1.1.4. Periksa kondisi MCB, jika terlihat ada yg rusak ganti MCB nya.

Check the condition of the MCB, if you see anything damaged, replace the MCB.

7.1.1.5. Periksa dan pastikan baut – baut pengikat skun terkait kencang.

Check and make sure the bolts securing the skun are tightened.

7.1.2. Perawatan pada unit *vacuum pump becker*.
Maintenance on the Becker vacuum pump unit

Gambar 2. Unit mesin vacuum untuk particle counter mesin filling bosch
Figure 2. Vacuum machine unit for Bosch filling machine particle counter

7.1.2.1. Bersihkan *body unit vacuum pump* yang berada di ruangan *vacuum pump*.
Cleaning body unit vacuum in the vacuum pump room.

7.1.2.2. Bersihkan filter udara pada unit *vacuum*.
Cleaning air filter in the unit vacuum.

7.1.2.3. Periksa kondisi *valve* – *valve* pada *header vacuum*, ganti jika *valve* terlihat rusak atau bocor.
Check the condition of the valves on the vacuum header, replace them if the valves appear damaged or leaking.

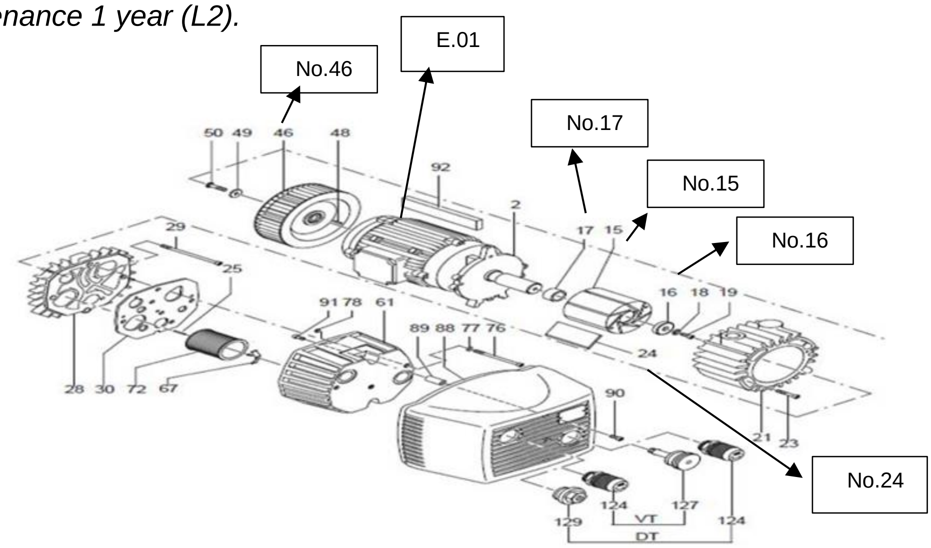
7.1.2.4. Periksa sambungan – sambungan pada selang mulai dari unit vacuum menuju ke ruangan mesin *filling bosch*.
Check the connections on the hose starting from the vacuum unit to the Bosch filling machine room.

7.1.2.5. Periksa sambungan pada *probe vacuum* yang berada di bagian dalam mesin *filling bosch*, kencangkan koneksinya jika terdapat yang kendur.
Check the connections on the vacuum probe inside the Bosch filling machine, tighten the connections if there are any loose ones.

7.1.2.6. Periksa tekanan vacuum yang terbaca pada *pressure gauge*, tekanan *vacuum* normal adalah – 0,70 sampai -0,80Mbar.
Check the vacuum pressure read on the pressure gauge, normal vacuum pressure is – 0.70 to -0.80 Mbar.

7.2. Perawatan 1 Tahunan (L4).

Maintenance 1 year (L2).



Gambar 3. *Vacuum Pump Becker*

Figure 3. Becker vacuum pump

7.2.1. Perawatan pada unit *vacuum pump becker.*

Maintenance on the Becker vacuum pump unit.

7.2.1.1. Periksa *bearing vacuum*, ganti *bearing* jika putaran vakum terdengan kasar diputar secara manual sedikit kasar, contoh bearing terdapar di nomer 16,17 dan bagian motor listriknya Lihat pada gambar 3. *Vacuum Pump Becker*. Point E.01

Check the vacuum bearing, replace the bearing if the vacuum rotation is roughly rotated manually a little rough, an example of the bearing is found at number 16, 17 and the electric motor part See Figure 3. Becker Vacuum Pump. Point E.01

7.2.1.2. Periksa kipas pendinginan pada *vacuum*, kencangkan jika kendor, contoh fan pendingin terdapat di nomer 46. Lihat pada gambar 3. *Vacuum Pump Becker*.

Check the cooling fan on the vacuum, tighten if loose, an example of a cooling fan is found at number 46. See Figure 3. Becker Vacuum Pump.

7.2.1.3. Periksa sambungan pada AS motor listrik ke bagian *housing carbon blade* kencangkan jika kendur atau ganti jika kondisinya terlihat kurang baik, contoh sambungan pada AS motor listrik ke bagian *housing carbon blade* terdapat di nomer 15. Lihat pada gambar 3. *Vacuum Pump Becker.*

Check the connection of the electric motor to the carbon blade housing tighten it if it is loose or replace it if the condition looks bad, an example of the connection of the electric motor to the carbon blade housing is number 15.

See Figure 3. Becker Vacuum Pump.

7.2.1.4. Periksa kondisi *carbon bladenya*, ganti jika kondisi *carbon blade* terlihat rusak atau ukurannya berubah dari ukuran standarnya, ukuran standar menyesuaikan pada jarak *housing carbon blade,* contoh *carbon blade* terdapat di nomer 24. Lihat pada gambar 3. *Vacuum Pump Becker.*
Check the condition of the carbon blade, replace it if the condition of the carbon blade looks damaged or the size changes from the standard size, the standard size adjusts to the carbon blade housing distance, an example of a carbon blade is found at number 24. See Figure 3. Becker Vacuum Pump.

8. REFERENSI

REFERENCE

Buku manual perawatan berkala Mesin Penghitung Partikel

Manual book of periodic maintenance Mesin Particle Counter

9. LAMPIRAN

ATTACHMENT

Formulir perawatan berkala terhadap Mesin *Particle Counter* SOP/EBI/EN-063-F01A.

Periodically Maintenance form of Particle Counter Machine *List No. SOP/EBI/EN-063-F01A.*

10. RIWAYAT

HISTORY

Tanggal ***Date***	**No. Change Control** ***Change Control No.***	**No. Revisi** ***No. Revision***	**Alasan Perubahan** ***Reason of Change***
	CC-EBI-EN-018-24	00	Dokumen baru *New Document*

11. DISTRIBUSI

DISTRIBUTION

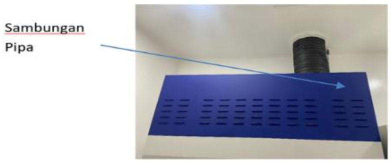
11.1. Departemen Engineering.
Engineering Department.

11.2. Departemen Produksi.
Production Departement

11.3. Departemen Sistem Mutu
Quality System Departement

etana Biotech **PT ETANA BIOTECHNOLOGIES INDONESIA**	**FORMULIR / *FORM*** **PERAWATAN TERHADAP MESIN FUME HOOD** ***MAINTENANCE OF FUME HOOD MACHINE***	Document No. : SOP/EBI/EN-063-F01A

Level	:
Month	:
Date	:
Day	:
Hours meter	:

Level	Component	Activity							Condition		Remark
		C	I	L	R	S	T	W	B	G	
L2											
	Rotation of the blower motor										
											
	Ampere on the blower motor										<2A
											
	Connection the blower pipe										
											
	1 phase electrical connection										
											
	Electrical installation										
											
	valve connection on the soft water supply										
											
	Hinges on the door										

etana Biotech PT ETANA BIOTECHNOLOGIES INDONESIA	**FORMULIR / *FORM*** **PERAWATAN TERHADAP MESIN FUME HOOD** ***MAINTENANCE OF FUME HOOD MACHINE***	Document No. : SOP/EBI/EN-063-F01A

Note :

No	Date	Start	Finish	Duration		Total personnel	Area	Remark
				Hour	Minute			

Remark : C : Cleaning (pembersihan) I : Inspection (pemeriksaan) L : Lubrication (pelumasan) R : Replacement (penggantian) S : Setting (penyetelan) T : Tightened (dikencangkan) W : Welding (pengelasan) G : Good (baik) B : Bad (jelek)	Level : L2 : Every 3 month	Done by : ( )	Checked by : ( )